Zenith Esports - Project Documentation

Version: 1.0.6.0
Developer: Taha Zaman
Date: May 5, 2026
GitHub Repository: tahazamancheema/zenithesports (https://github.com/tahazamancheema/zenithesports)

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1. Introduction

Zenith Esports is a premium tournament management platform designed for competitive gaming, specifically tailored for mobile titles like PUBG Mobile. It provides a seamless experience for players to discover tournaments, register teams, and for administrators to manage the competitive lifecycle.

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2. System Architecture (Component Relations)

The application follows a modular React architecture with centralized state management via Context API and real-time data persistence through Supabase.

2.1 Component Tree (Class Relations)

In the React context, "Class Relations" are represented by the component hierarchy and data flow.

```
graph TD
    App --> AuthProvider
    AuthProvider --> ErrorBoundary
    ErrorBoundary --> AppLayout
    AppLayout --> Navbar
    AppLayout --> MainContent[Routes/Pages]
    AppLayout --> Footer

    subgraph Pages
        MainContent --> LandingPage
        MainContent --> TournamentsPage
        MainContent --> TournamentDetailsPage
        MainContent --> AuthPage
        MainContent --> ProfilePage
        MainContent --> AdminDashboard
        MainContent --> RegistrationPage
    end

    subgraph Hooks
        AuthPage -.-> useAuth
        ProfilePage -.-> useAuth
        RegistrationPage -.-> useRegistration
        TournamentsPage -.-> useTournaments
        AdminDashboard -.-> useSupabaseDB
    end
```

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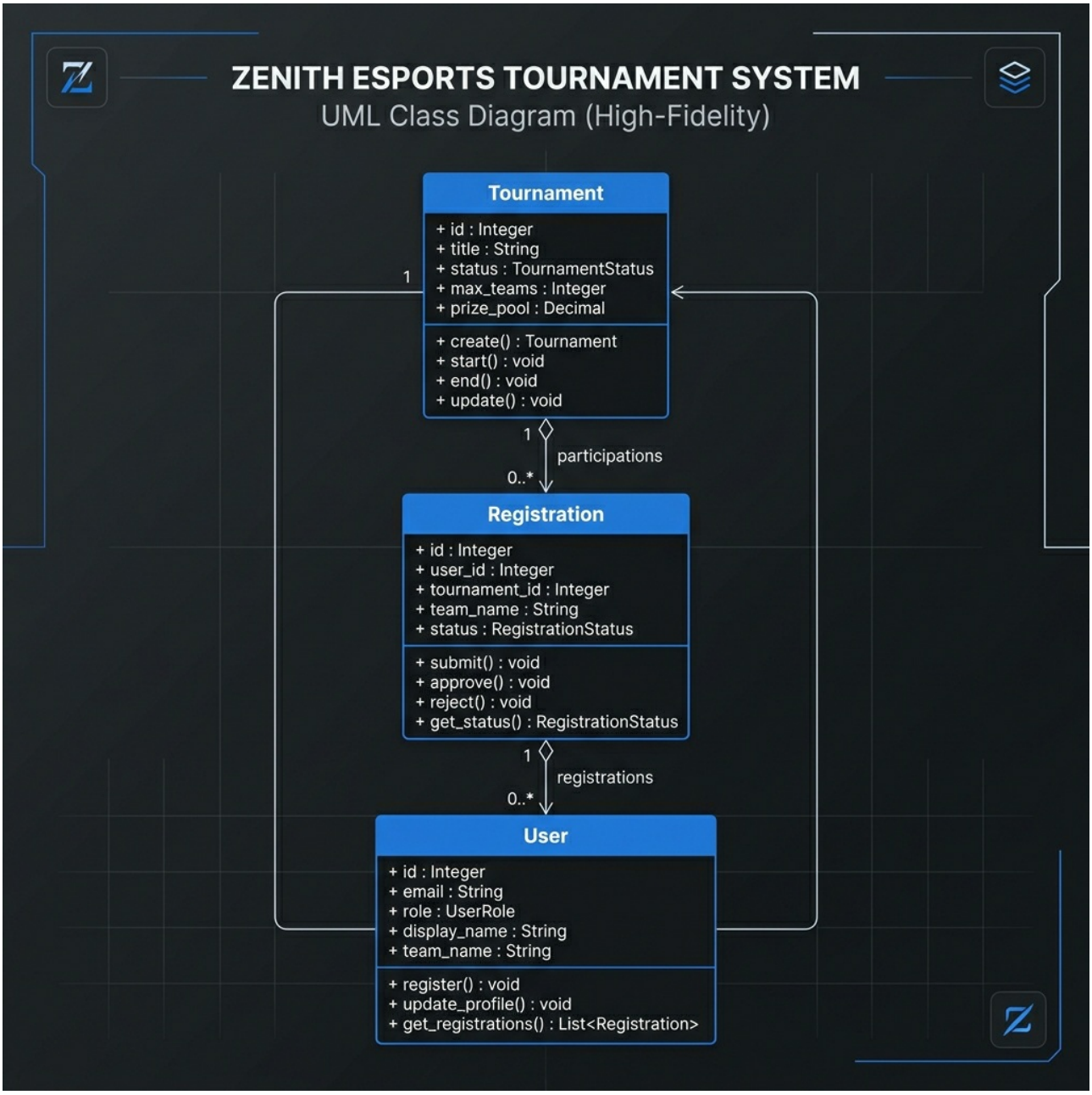
3. Data Model (UML Class Diagram)

The following UML Class Diagram represents the core data entities and their relationships within the Supabase PostgreSQL database.



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4. Activity Diagram: Tournament Registration

This diagram illustrates the logical flow of a user registering for a competitive event, including authentication and admin review.

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5. Use Cases

UC-01: Team Registration

- **Actor:** Player (Captain)
- **Description:** The captain registers their team for an upcoming tournament by providing player details and social media verification screenshots.
- **Pre-condition:** User must be authenticated.
- **Post-condition:** A registration record is created with a 'pending' status.

UC-02: Tournament Management

- **Actor:** Administrator
- **Description:** The admin creates new tournaments, sets prize pools, and manages registration deadlines.
- **Pre-condition:** User must have the 'admin' role.
- **Post-condition:** Tournament details are updated in the database and visible to all users.

UC-03: Competitive Eligibility Verification

- **Actor:** System
- **Description:** The system automatically checks if players listed in a new registration are already registered in the same

tournament under a different team to prevent double-entry.
- **Pre-condition:** Registration form is submitted.
- **Post-condition:** Registration is either allowed or blocked based on data integrity rules.

UC-04: Profile Management

- **Actor:** Player
- **Description:** A player updates their permanent team profile (logo, member IGNs). This data automatically populates future registration forms to speed up the process.
- **Pre-condition:** User is logged in.
- **Post-condition:** User record is updated in the `users` table.

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6. Version History

Version	Date	Changes	Developer
1.0.0.0	2026-04-20	Initial Project Architecture & Supabase Setup	Taha Zaman
1.0.2.0	2026-04-25	Real-time Tournament Data Synchronization	Taha Zaman
1.0.4.0	2026-04-30	Landing Page Redesign & UI Performance Optimization	Taha Zaman
1.0.5.0	2026-05-02	Tournament Registration Workflow & Duplicate Checks	Taha Zaman
1.0.6.0	2026-05-05	Final Polish, Documentation, and Stability Fixes	Taha Zaman